

Shivam Sharma

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Summary — Experienced in designing, developing, and maintaining high-performance software systems using modern technologies and agile methodologies. Proven track record in delivering scalable solutions and collaborating effectively in cross-functional teams. Strong analytical and problem-solving skills, with a focus on innovation and quality-driven results.

Skills

Databases MongoDB, MySQL, PostgreSQL, DynamoDB
Cloud AWS
Languages Python, JavaScript (ES6+), TypeScript, Java
Frontend ReactJS, Angular

Backend Node.js, Express.js, SpringBoot
Architecture RESTful, Serverless Application Model (SAM),
Microservices, event-driven architecture

Experience

BMW Techworks India

May 2025 – Present

Cloud Developer - Senior Associate

Project: CAIP(Connected AI Platform)

- **Backend Development** — Designing and implementing robust server-side solutions using **NestJS** and **MongoDB**, ensuring scalability, maintainability, and high performance.
- **System Architecture** — Actively contributing to **end-to-end backend architecture design**, including **modular service-oriented structures**, and engineering a **highly-available AWS deployment pipeline** for automated, reliable releases.
- **Data Modeling** — Creating and optimizing **document-based database schemas** tailored for high-volume, complex data operations, with a focus on query performance and maintainability.
- **Security & Authorization** — Architecting and implementing **enterprise-grade Role-Based Access Control (RBAC)**, along with **multi-token authentication mechanisms** (access/refresh tokens, JWT, API keys) to enforce fine-grained authorization.
- **API Engineering** — Developing **RESTful APIs** for diverse business domains such as **user lifecycle management** and **space resource management**, integrating **validation, error handling, and performance optimization** best practices.
Project: OMS (Order Management System)
- **Distributed Microservices Design** — Architected and built a production-ready, **event-driven Order Management System** as a **NestJS monorepo** comprising a REST API service and a decoupled async worker service, enabling independently scalable and resilient processing pipelines.
- **Event-Driven Messaging** — Designed a **dual-queue strategy** supporting **BullMQ** (Redis-backed) and **AWS SQS** via a pluggable producer/consumer abstraction, allowing zero-code provider switching through environment configuration and ensuring reliable, at-least-once message delivery with exponential backoff retry.
- **Order State Machine & Action Registry** — Engineered a central **Action Registry** mapping business scopes to ordered sync/async processing steps, backed by a strict **finite-state-machine** (RECEIVED → IN_PROGRESS → AWAITING_CALLBACK → COMPLETED/FAILED), enabling resumable async workflows with **step-index persistence** so the worker resumes from the exact checkpoint after external callbacks without re-executing completed steps.
- **Security & Policy-as-Code** — Integrated **Open Policy Agent (OPA)** for fine-grained authorization, implementing custom NestJS guards with **scope-driven permission evaluation** from JWT bearer tokens, supporting both role-based and body-scope-derived access control across all API endpoints.
- **Observability & Structured Logging** — Built a centralized, **structured logging service** with **CDLL-compliant output formatting**, propagating rich contextual metadata (orderId, spaceId, scope, callerType) across all service layers to support **production-grade observability** and operational diagnostics.
- **Kubernetes & Helm Deployment** — Authored **Helm charts** for Kubernetes deployment of both API and worker services, including **hardened security contexts** (non-root containers, dropped Linux capabilities, seccomp profiles), ConfigMap-driven configuration, and environment-specific value overrides for test and production.
- **Resilient Data Modeling** — Designed **MongoDB schemas** for Orders and Order Timelines with indexed fields for high-throughput queries and maintained a **full audit trail** of all status transitions, supporting compliance and operational debugging.

Mercedes-Benz Research and Development India

April 2023 – May 2025

Full Stack Engineer (Technical Consultant)

Project: ACID (Automatic Check for Daily and Initial File)

- Developed and deployed a Python Django application to upload and process XML files with multi-level hierarchical validation based on predefined business rules.
- Implemented efficient service logic in Python for XML parsing, data validation, and Excel report generation using openpyxl, pandas, and lxml.etree.

- Optimized file processing performance, handling large XML files (up to 100MB) with reduced memory consumption through batch processing and efficient garbage collection.

- Debugged and Migrated the application from an on-premise environment to AWS Cloud using AWS CDK in a lift-and-shift approach, ensuring minimal downtime and seamless transition.

Project : User Access Management (UAM)

- Collaborated with the team to implement role-based access control in UAM, designing APIs in Spring Boot to handle dynamic user roles and privileges, ensuring seamless integration with dependent applications.
- Developed Liquibase migration scripts for PostgreSQL to support new application features, including dynamic privilege assignments, ensuring zero downtime during database updates.
- Worked on distributed caching strategies using Hazelcast, enabling faster access to frequently used user and role data.

Project: iTrans (Translation management application)

- Participated in low-level and high-level architectural design discussions, driving quality, scalability, and engineering excellence for complex applications.
- Designed and implemented high-performance, scalable RESTful APIs using Node.js within a microservice architecture, ensuring seamless integration and responsiveness and wrote tests using Jest.
- Leveraged AWS SAM (Serverless Application Model) to define, build, and deploy serverless applications, enhancing scalability and reducing deployment times.
- Utilized a variety of databases to handle relational and JSON-based storage requirements effectively.
- Developed and maintained CI/CD pipelines with tools such as GitHub Actions, automating build, test, and deployment processes to ensure consistent delivery and quality assurance.
- Optimized database performance through indexing and query tuning techniques
- Enabled cross-functional collaboration between development and product teams, ensuring clear alignment with business requirements and project timelines.

alphastream.ai

Feb 2022 – April 2023

Member of Technical Staff 1

- Implemented 50+ user-centric UI features, including context menus, navigation enhancements, hooks, and annotation tools, elevating user experience and engagement.
- Engineered responsive, functional components leveraging ReactJS concepts, ensuring high performance and seamless scalability for web application UIs.
- Architected and optimized frontend structures, with a focus on grid systems, collaborating closely with team members to establish robust and maintainable UI architectures.
- Maintained and enhanced the Gateway module in Flask of a data extraction pipeline, integrating advanced functionalities such as metadata modification, secure SFTP operations, and RESTful API interactions of critical Finance Domain.
- Designed, developed, and deployed a full-stack text summarizer using Python, applying end-to-end development principles to deliver a high-impact, production-ready solution.

Education

Jaypee University Of Engineering and Technology

GPA : 7.2

Bachelor of Technology in Computer Science and Engineering

2018-2022

Certifications

- AWS - Certified Cloud Practitioner